

K9HZ 100W LPF – FILTER BOARD
BUILD INSTRUCTIONS for PCB V1.10
JUNE 11, 2026

The K9HZ LPF board set is intended to be a very compact yet effective 11 band low-pass filter network to meet FCC regulations. While this board was intended to be use in a set of three, it can be used stand-alone for just using the filters.

Operating Data

Power Requirements: 12-15 VDC at 100ma max (provided by the Control Board).
Frequency Range: 1.8MHz – 54MHz/ 160M – 6M in 11 bands inclusive.
RF Input: 150W Maximum. Input and Output with bypass.
Control: BCD band data from the Control Board via two connectors.
Performance: N-th order harmonic reduction more than than 43 db.

Inventory and Prework

Spend time to determine if you want all 11 bands for 1.8MHz-54MHz/ 160M-6M (all amateur bands included in that frequency range). If you wish to delete a particular band or build that filter later, just remove those parts from the BOM before ordering. The 100W BOM is:

Qty	Part Value	Part Reference
1	14pF 200V	C70
5	27pF 200V	C63, C69, C39A, C67A, C100A
4	33pF 200V	C65, C39B, C67B, C100B
1	39 pF 200V	C71
3	47pF 200V	C60, C62, C64
2	56pF 200V	C57, C75
	60pF 200V	C39, C67, C100 (made from above A and B parts stacked)
2	68pF 200V	C66, C68
8	82pF 200V	C24, C35, C41, C53, C72, C85, C90, C105
2	91pF 200V	C51, C82
9	100pF 200V	C32, C38, C52, C56, C58, C74, C76, C94, C99
2	120pF 200V	C37, C98
11	150pF 200V	C27, C28, C33, C36, C42, C61, C83, C95, C97, C102, C104
2	180pF 200V	C50, C81
4	220pF 200V	C34, C49, C80, C96
13	330pF 200V	C25, C43, C44, C48, C54, C55, C59, C73, C77, C79, C87, C88, C103
2	390pF 200V	C26, C101
6	470pF 200V	C30, C40, C47, C78, C89, C92
2	560pF 200V	C31, C93
2	1200pF 200V	C46, C86
2	1500pF 200V	C29, C91
2	2200pF 200V	C45, C84

24	0.1uF 50V	C106, C107, C108, C109, C110, C111, C112, C113, C114, C115, C116, C117, C118, C119, C120, C121, C122, C123, C124, C125, C126, C127, C157, C158	
2	RF interconnect	J17, J22	
1	IDX2x5 Male Box Connector	J26, J27	
24	HF41F-12-ZS	K8, K9, K10, K11, K12, K13, K14, K15, K16, K17, K18, K19, K20, K21, K22, K23, K24, K25, K26, K27, K28, K29, K32, K33	Surplus from eBay
1	0.170uH	L24	8 Turns of #20 Wire on a T68-17 Core.
2	0.188uH	L13, L35	8 Turns of #20 Wire on a T68-17 Core.
1	0.305uH	L23	7 Turns of #20 Wire on a T68-6 Core.
2	0.335uH	L12, L34	7 Turns of #20 Wire on a T68-6 Core.
2	0.337uH	L11, L33	7 Turns of #20 Wire on a T68-6 Core.
1	0.342uH	L22	7 Turns of #20 Wire on a T68-6 Core.
1	0.406uH	L21	8 Turns of #20 Wire on a T68-6 Core.
2	0.447uH	L10, L32	9 Turns of #20 Wire on a T68-6 Core.
1	0.471uH	L20	9 Turns of #20 Wire on a T68-6 Core.
2	0.519uH	L9, L31	10 Turns of #20 Wire on a T68-6 Core.
1	0.608uH	L19	10 Turns of #20 Wire on a T68-6 Core.
2	0.670uH	L8, L30	11 Turns of #20 Wire on a T68-6 Core.
1	0.843uH	L18	12 Turns of #20 Wire on a T68-6 Core.
2	0.928uH	L7, L29	13 Turns of #20 Wire on a T68-6 Core.
1	1.22uH	L17	15 Turns of #20 Wire on a T68-6 Core.
2	1.34uH	L6, L28	16 Turns of #20 Wire on a T68-6 Core.
1	1.60uH	L16	16 Turns of #20 Wire on a T68-2 Core.
2	1.76uH	L5, L27	17 Turns of #20 Wire on a T68-2 Core.
1	2.43uH	L15	20 Turns of #20 Wire on a T68-2 Core.
2	2.68uH	L4, L26	21 Turns of #20 Wire on a T68-2 Core.
1	4.73uH	L14	28 Turns of #20 Wire on a T68-2 Core.
2	5.21uH	L3, L25	30 Turns of #20 Wire on a T68-2 Core.
3	T68-17 Cores	For L3-L35	KitsAndParts.com T41 V012.6 LPF Kit
21	T68-6 Cores	For L3-L35	KitsAndParts.com T41 V012.6 LPF Kit
9	T68-2 Cores	For L3-L35	KitsAndParts.com T41 V012.6 LPF Kit
38	Feet #20 Transformer Wire	For L3-L35	KitsAndParts.com T41 V012.6 LPF Kit

The filter board can also be built as a 20W maximum board for QRP operations. The BOM for that combination of parts is listed in the BOM directory.

Keep the capacitors in their envelopes until you need them as they are hard to discern if you mix them up. A good 45W soldering pencil, solder, flux, and a good set of forceps are all that is needed to complete this board. Begin by wiping the board, front and back, with Isopropyl alcohol. This will make soldering parts on the board easier.

Building the LPF Filter board – Pre Assembly Tasks

Spend time winding the torroids before building the board. If you have an LCR meter or a VNA, wind each inductor... measure it, and label it. The number of turns given in the BOM is a very good starting point and should work sufficiently as they have been verified in our quality control process. Spread the windings evenly over the cores. If you don't have a way to measure the inductance of the wound inductor, stay with the number of turns suggested in the BOM. Wind the cores only for the bands you want to use with your finished radio (**be warned that it may be difficult to add the inductors, capacitors, and relays, for bands omitted in the initial build**).

The following is a table that associates the inductors, capacitors, and relays to each associated band. You don't have to install parts in this table for the bands you never intend to use (that is, omit them!). The bypass position (K32, K33) will be used when no filter is in that band position, programmable in software.

BAND	Inductors	Capacitors	Relays
160	L3, L14, L25	C28, C29, C45, C55, C59, C60, C77, C84, C91, C102	K16, K21
80	L4, L15, L26	C26, C30, C46, C56, C61, C62, C76, C86, C92, C101	K14, K19
60	L5, L16, L27	C31, C40, C44, C57, C66, C67, C75, C87, C89, C93	K13, K20
40	L6, L17, L28	C25, C32, C43, C54, C58, C73, C88, C94, C103	K12, K22
30	L7, L18, L29	C27, C33, C47, C68, C78, C95, C104	K17, K23
20	L8, L19, L30	C34, C38, C64, C79, C96	K15, K24
17	L9, L20, L31	C24, C35, C42, C52, C71, C74, C83, C90, C105	K9, K25
15	L10, L21, L32	C36, C49, C65, C89, C97	K8, K26
12	L11, L22, L33	C37, C50, C63, C81, C98	K10, K27
10	L12, L23, L34	C38, C41, C53, C69, C72, C85, C99	K11, K28
6	L13, L24, L35	C39, C51, C70, C82, C100	K18, K29

Sequence of Component Placement

The order of component placement on the LPF filter board is critical.

1. All capacitors (except those you have determined are associated with bands which you will never transmit.. as from the table above) should be populated **FIRST**. Capacitors are in **BOTH** sides of the board.
2. Next, start in the **CENTER** of the board and alternate placing **RELAYS ON THE BOTTOM** and **INDUCTORS ON THE TOP** in a circle expanding out to the edges to allow soldering the relays and inductors. There is no exact way to do this, but it will become obvious how to proceed as you progress. The relays are harder to place and solder once the inductors are in place, so place a few relays first. Leave relays **K32** and **K33** for the next step.
3. Once relays (except for **K32** and **K33**) and inductors are soldered in place, place and solder all connectors in place.
4. Finally, solder **K32** and **K33** in place.
5. This concludes building the LPF Filter module. Clean the board with IPA.
6. The Filter Board can be tested using the test algorithms provided on the software GITHUB.